Assignment of FODS

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#1. Logic Design: Create a flowchart that illustrates the logic for a Leap
Year Checker. The diagram must clearly show the input of a year, the
conditional processing steps, and the final output.

year = int(input("Enter a year: "))

if year % 400 == 0:
   print(year, "is a Leap Year")
elif year % 100 == 0:
   print(year, "is not a Leap Year")
elif year % 4 == 0:
   print(year, "is a Leap Year")
else:
   print(year, "is not a Leap Year")
#2. Unit Conversion: Write a program that takes a temperature input in Celsius
and converts it into Fahrenheit, Kelvin, and Rankine. Display all results in a
clear, formatted table.

celsius = float(input("Enter temperature in Celsius: "))

fahrenheit = (celsius * 9/5) + 32
kelvin = celsius + 273.15
rankine = (celsius + 273.15) * 9/5

print("\nTemperature Conversion")
print("-------------------------")
print("Celsius     =", celsius)
print("Fahrenheit  =", fahrenheit)
print("Kelvin      =", kelvin)
print("Rankine     =", rankine)
# 3. Coordinate Geometry: Write a program that prompts the user to enter the
(x, y)
# coordinates of a point. Determine and display which quadrant the point lies
in (1st,
# 2nd, 3rd, or 4th), or if it lies on an axis or the origin.

x = float(input("Enter x coordinate: "))
y = float(input("Enter y coordinate: "))

if x > 0 and y > 0:
   print("Point lies in 1st Quadrant")
elif x < 0 and y > 0:
   print("Point lies in 2nd Quadrant")
elif x < 0 and y < 0:
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   print("Point lies in 3rd Quadrant")
elif x > 0 and y < 0:
   print("Point lies in 4th Quadrant")
elif x == 0 and y == 0:
   print("Point lies at Origin")
elif x == 0:
   print("Point lies on Y-axis")
else:
   print("Point lies on X-axis")

# 4. Advanced Math: Write a program that takes a radius as input and
calculates the
# following for a sphere: Diameter, Circumference, Surface Area, and Volume.
# Additionally, calculate the natural log of the volume.

import math

radius = float(input("Enter radius of sphere: "))

diameter = 2 * radius
circumference = 2 * math.pi * radius
surface_area = 4 * math.pi * radius**2
volume = (4/3) * math.pi * radius**3
log_volume = math.log(volume)

print("\nSphere Calculations")
print("-----------------------")
print("Diameter =", diameter)
print("Circumference =", circumference)
print("Surface Area =", surface_area)
print("Volume =", volume)
print("Natural Log of Volume =", log_volume)
# 5. Salary &amp;amp; Tax System: Create a program to input an employee&amp;#39;s
gross salary.
# Calculate the net salary after applying the following tax rules:
# Tax-Free: Below £12,500
# 20% Tax: £12,501 - £50,000
# 40% Tax: £50,001 - £125,000
# 45% Tax: Above £125,000

salary = float(input("Enter gross salary: "))

if salary < 12500:
   tax = 0
elif salary <= 50000:
   tax = salary * 0.20
elif salary <= 125000:
   tax = salary * 0.40
else:
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    tax = salary * 0.45

net_salary = salary - tax

print("\nSalary Details")
print("--------------------")
print("Gross Salary =", salary)
print("Tax Amount =", tax)
print("Net Salary =", net_salary)
# 6. Fibonacci Sequence: Write a program to display the Fibonacci sequence up
to a
# user-defined number of terms. The program should also output the sum of the
# sequence generated.

terms = int(input("Enter number of terms: "))

a = 0
b = 1
sum_series = 0

print("Fibonacci Sequence:")

for i in range(terms):
    print(a, end=" ")
    sum_series += a
    c = a + b
    a = b
    b = c

print("\nSum of sequence =", sum_series)
# 7. Pattern Filtering: Write a program to find all numbers between 500 and
1500
# (inclusive) that are multiples of 7 but not multiples of 5. Allow the user
to define
# their own custom range for this search.

start = int(input("Enter start range: "))
end = int(input("Enter end range: "))

print("Numbers divisible by 7 but not by 5:")

for number in range(start, end + 1):
    if number % 7 == 0 and number % 5 != 0:
        print(number, end=" ")
# 8. Input Validation (Palindrome): Write a program that asks the user to
enter a 5-digit
# positive integer. Validate the input to ensure it is exactly 5 digits and
positive. If
# valid, determine if the number is a palindrome (reads the same forwards and
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# backwards).

number = input("Enter a 5-digit positive integer: ")

if number.isdigit() and len(number) == 5:
   if number == number[::-1]:
       print("The number is a palindrome.")
   else:
       print("The number is not a palindrome.")
else:
   print("Invalid input. Enter exactly 5 digits.")
# 9. Inventory Menu: Create a menu-driven program for a simple shop. The menu
should
# allow the user to:
# a. Add an item price to a running total.
# b. Apply a 10% discount to the current total.
# c. View the final total and exit.
# d. Ensure the program loops until the user chooses to exit.

total = 0

while True:
   print("\nMenu")
   print("1. Add Item Price")
   print("2. Apply 10% Discount")
   print("3. View Total and Exit")

   choice = input("Enter your choice: ")

   if choice == "1":
       price = float(input("Enter item price: "))
       total += price
       print("Current Total =", total)

   elif choice == "2":
       total = total - (total * 0.10)
       print("Discount Applied")
       print("New Total =", total)

   elif choice == "3":
       print("Final Total =", total)
       print("Exiting Program...")
       break

   else:
       print("Invalid choice")
# 10. Dice Rolling Game: Build a game where the computer and the user each
roll a
# random 6-sided die for 5 rounds.
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# a. Track the score for both.
# b. After each round, announce the winner of that round.
# c. At the end of 5 rounds, declare the overall winner or if it was a draw.

import random

user_score = 0
computer_score = 0

for round in range(1, 6):
    print("\nRound", round)

    user_roll = random.randint(1, 6)
    computer_roll = random.randint(1, 6)

    print("User rolled:", user_roll)
    print("Computer rolled:", computer_roll)

    if user_roll > computer_roll:
        print("User wins this round")
        user_score += 1

    elif computer_roll > user_roll:
        print("Computer wins this round")
        computer_score += 1

    else:
        print("Round Draw")

print("\nFinal Scores")
print("User Score =", user_score)
print("Computer Score =", computer_score)

if user_score > computer_score:
    print("User is the Overall Winner!")
elif computer_score > user_score:
    print("Computer is the Overall Winner!")
else:
    print("The Game is a Draw!")
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